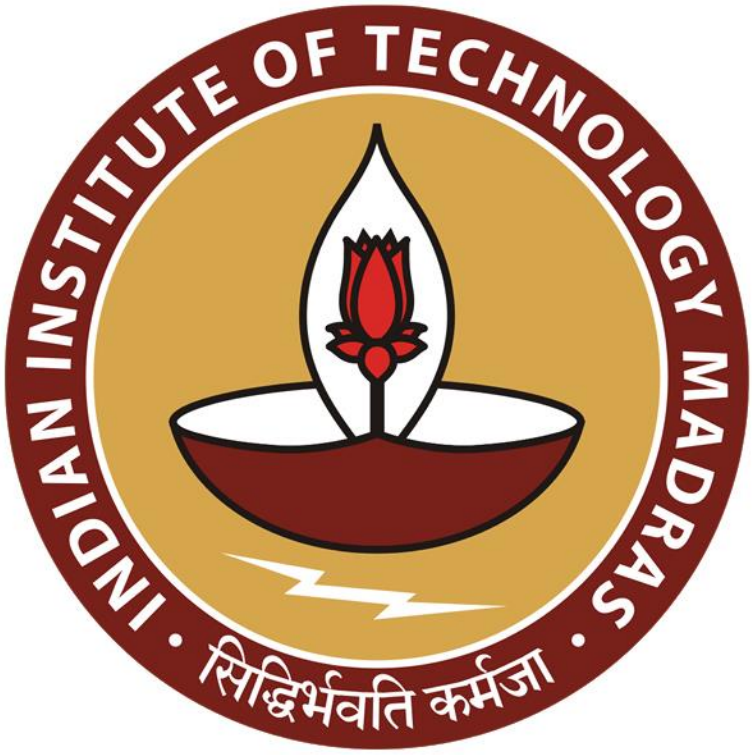


IIT Madras

ONLINE DEGREE

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Principal Component Analysis (continued)

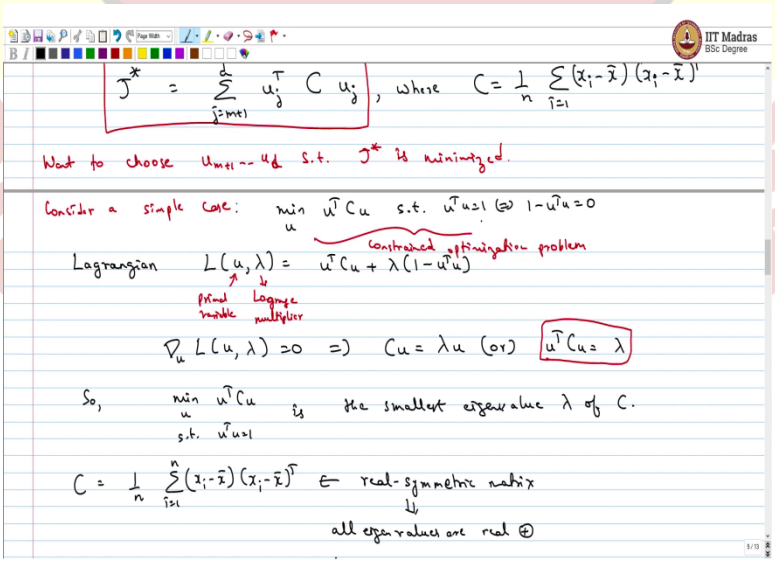
Welcome to lecture 4 in week 6 of machine learning foundations. We are going to continue the discussion of PCA, principal component analysis. And since this is an involved topic, we will split this a into two lectures.

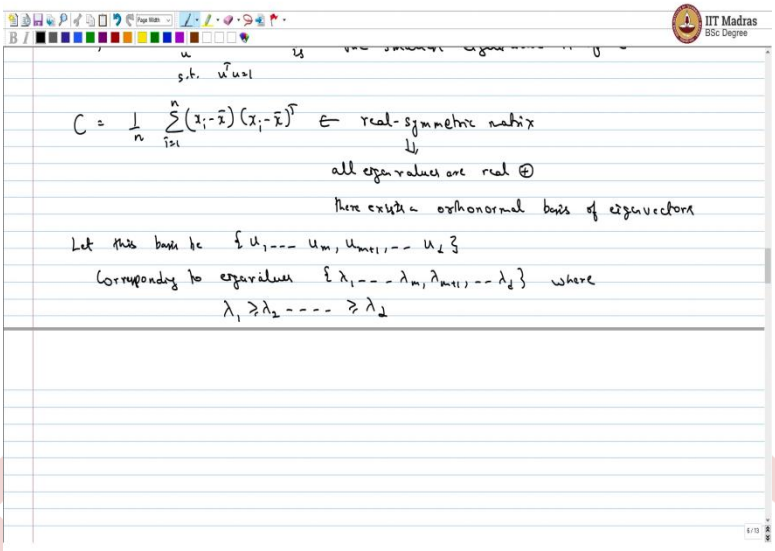
In the last lecture, we started with, for a given data set and a given input parameter, which signifies the lower, which signifies the dimension of a lower, lower dimensional subspace, we defined what reconstruction error is, and we are trying to figure how to find the best lower dimensional subspace to project the data onto, so that the reconstruction error is minimised.

In this lecture, we will continue this discussion and wrap up and arrive at the algorithm, for doing this dimensionality reduction, the PCA way and that will involve picking some eigenvalues and eigenvectors of a matrix for certain matrix and picking a subset of them and using that to do, work out the projected data points. We will look at that in detail today.

After that, we will look at an example to see all these ideas in action, which is in particular, for a very simple example, we will see how to apply the PCA algorithm to understand all the main ideas underlying PCA algorithm.

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Consider a simple case to understand how to choose u's, before that want to choose u_{m+1} +…..+u_d such that J^* is minimised. That is the goal. To do this, I am going to take a much simpler case, consider the following simple case. Minimum, just one u^TCu such that $u^Tu = 1$.

Now I am going to show you a tool but I will try to do some explicit calculation using calculus also. But the most general approach for such constraints, so this is what is called as a constrained problem, constrained optimization problem. The way to go about solving this is by using what is called a Lagrangian.

Lagrangian $L(u,\lambda) = u^TCu + \lambda (1 - u^Tu)$ it has two parameters, u is primal variable and is called the Lagrange multiplier, this is the two variable and this is called a Lagrange multiplier. So, what you do is this, you take the objective u^TCu, so, you convert this constrained optimization problem into something that is unconstrained by throwing in what is called a Lagrange multiplier.

This is what you do, such that $u^Tu = 1$ or now this constraint can be written as $1 - u^Tu=0$. And that is the constraint for writing down the Lagrangian has to be in the form of either equal to 0 or it should be an inequality constraint. And that is the reason why we take this form.

Now, what you have to do is, you have to take the Lagrangian and differentiate it with respect to u and equate it to 0. This implies, what you get? This implies $Cu = \lambda u$ or $u^TCu = \lambda$ and that is really the solution. So, min over u^TCu (such that $u^Tu =1$)is λ. Now you want to minimise this, so just pick the smallest λ.

Now, the important factor is let us understand this. So, C is this matrix if you recall and this is a real symmetric matrix. What does that mean? Which in turn implies all eigenvalues are real

+ there exists a orthonormal, is the smallest eigen value λ of C. So, that is the minimiser, u transpose Cu equals λ and you just have to pick this smaller sets, I mean understand this. So, maybe I have to do some explanation.

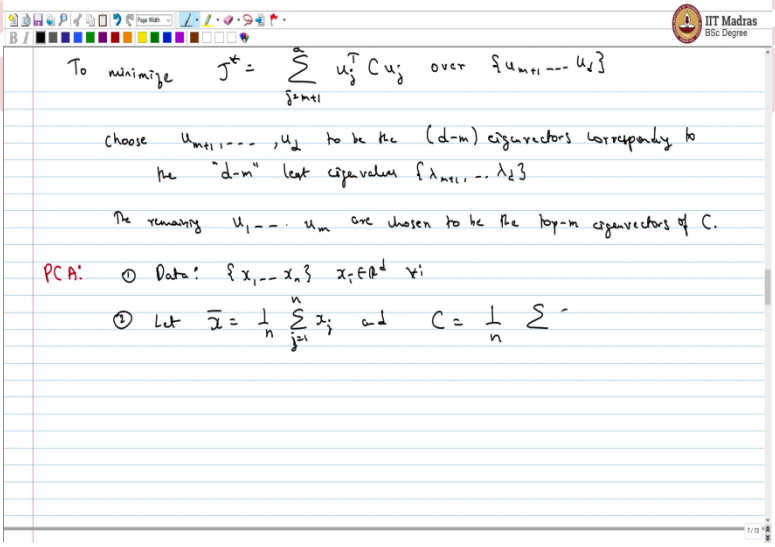
So, we just figured that if you see here, we got $u^TCu = \lambda$. And so, when I want to minimise this, when I want to find the smallest such u^Tu, when I do that minimise for u^TCu, then I have to just pick the smallest λ. So, the λ is an eigen value of C, so take all the eigen values of C and pick the smallest one and that could minimise u^TCu, solve the rest out.

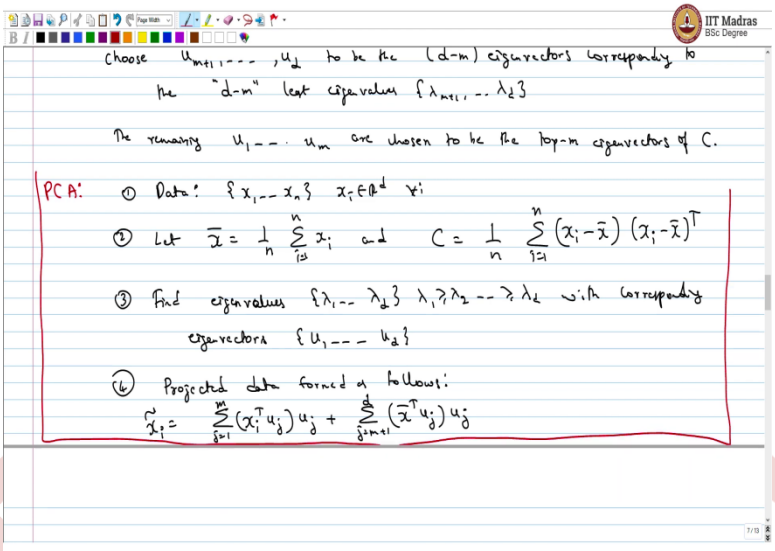
Now we come back to the original matrix C. So, the original matrix C in the case of PCA is this, and there exists an orthonormal basis of eigenvectors. Let this, now, what is C, what is the dimension, it is a d × d matrix basis be {u1……. till u_m, u_{m+1} +…..+u_d) corresponding to eigen values {λ1…… till λm , λ_{m+1}+…..+ λ_d}, where $\lambda_1 \geq \lambda_{2\ldots\ldots}$ and so on until λ_d. So, λ_1 is the largest eigenvalue and λ_d is smallest.

Now, if you want to, so, just to recap this whole thing again and just follow this train of thought, logic is that you are trying to minimise J^* equals whatever is highlighted in this red box. And if I take a much simpler case, where there is just one term and I want to minimise, here, $u^TCu = \lambda$ as I have highlighted here , the constrained optimization problem, then it turns out to be the smallest eigenvalue of C.

Now, the way to go about this is so, if I want to minimise this, I take the smallest eigen value and the corresponding eigen vector out of the equation and I can extend this logic to not greater than 1 also.

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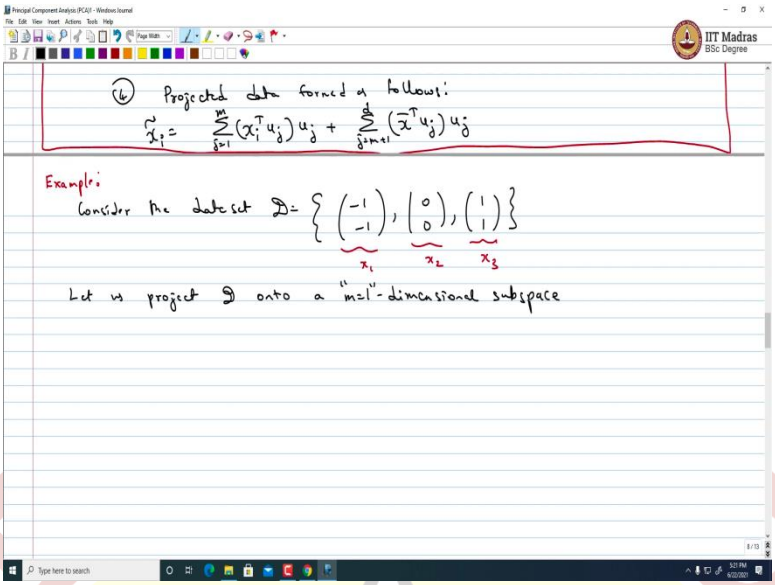


So, coming back to, minimise, the main point of this is to minimise $J^* = \sum_{j=m+1}^{d} u_j^T C u_j$ .All you have to do is choose u_{m+1},…..,ud. to be the (d – m) eigen vectors corresponding to, to the (d – m) least eigen values, just say λ_{m+1},…….. until λ_d.

Now the remaining $u_{1+\ldots\ldots\ldots+}$ until um chosen to be the top m eigenvectors of this matrix C. So, that is really the PCA logic. So, to summarise, what do you do with PCA? PCA is given data x_1…… to xn, $x_1 \in \mathbb{R}^d$ $\forall$ i. what you do is choose from, let from the average point $\bar{x} = \frac{1}{n}\sum_{j=1}^{n}(x_j)$ and so form this, form the matrix $C = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^T$.

Next step, find eigen values $\{\lambda_1 \ldots . \lambda d\}$, $\lambda_1 \geq \lambda_{2\ldots\ldots\ldots}$ and so on until λ_d with corresponding eigen vectors, say $\{u_1 \ldots\ldots$ until $u_d\}$, this is what you do. next find the projected data as follows, projected data as follows xi tilde equals $\sum_{j=1}^{m}(x_i^T u_j)u_j + \sum_{j=m+1}^{d}(x_i^T u_j)u_j$. So, this is the PCA algorithm, if you want to call this and that minimises this reconstruction error, which is denoted by J^*. Let us do a simple example to illustrate the main ideas behind PCA, the algorithm itself.

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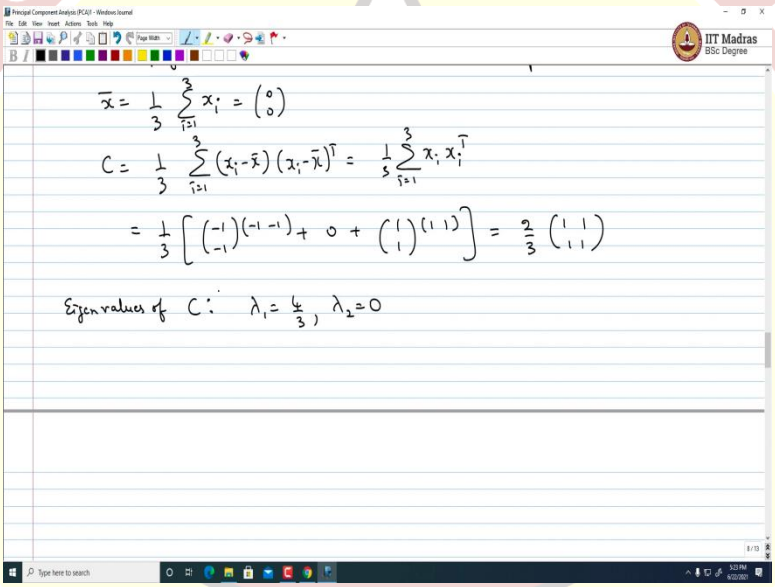
⑥ Projected data formed as follows:

$$\tilde{x}_i = \sum_{j=1}^{m} (x_i^T u_j) u_j + \sum_{j=m+1}^{d} (\bar{x}^T u_j) u_j$$

Example:

Consider the dataset $D = \left\{ \underbrace{\begin{pmatrix} -1 \\ -1 \end{pmatrix}}_{x_1}, \underbrace{\begin{pmatrix} 0 \\ 0 \end{pmatrix}}_{x_2}, \underbrace{\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{x_3} \right\}$

Let us project D onto a "m=1"-dimensional subspace

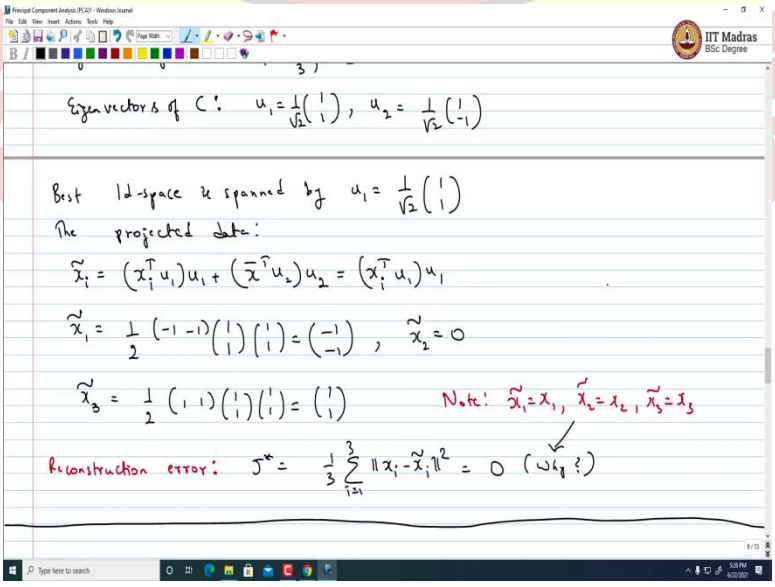


$$\bar{x} = \frac{1}{3} \sum_{i=1}^{3} x_i = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$C = \frac{1}{3} \sum_{i=1}^{3} (x_i - \bar{x})(x_i - \bar{x})^T = \frac{1}{3} \sum_{i=1}^{3} x_i x_i^T$$

$$= \frac{1}{3} \left[\begin{pmatrix} -1 \\ -1 \end{pmatrix} (-1 \; -1) + 0 + \begin{pmatrix} 1 \\ 1 \end{pmatrix} (1 \; 1) \right] = \frac{2}{3} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

Eigenvalues of C: $\lambda_1 = \frac{4}{3}$, $\lambda_2 = 0$



Eigenvectors of C: $u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $u_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Best 1d-space is spanned by $u_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

The projected data:

$$\tilde{x}_i = (x_i^T u_1) u_1 + (\bar{x}^T u_2) u_2 = (x_i^T u_1) u_1$$

$$\tilde{x}_1 = \frac{1}{2} (-1 \; -1) \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \quad \tilde{x}_2 = 0$$

$$\tilde{x}_3 = \frac{1}{2} (1 \; 1) \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Note: $\tilde{x}_1 = x_1$, $\tilde{x}_2 = x_2$, $\tilde{x}_3 = x_3$

Reconstruction error: $J^* = \frac{1}{3} \sum_{i=1}^{3} \| x_i - \tilde{x}_i \|^2 = 0$ (why?)

So, consider the data, it is a simple example. You have data like this $\left\{\begin{pmatrix}-1\\-1\end{pmatrix}, \begin{pmatrix}0\\0\end{pmatrix}, \begin{pmatrix}1\\1\end{pmatrix}\right\}$. Let us call this x1, x2 and x3. Now let us project D onto a m = 1 dimensional subspace. So, to go about this, let us first understand what is $\bar{x}$ which is $\frac{1}{3}\sum_{i=1}^{3}(x_i)$. Given this data it is easy to see that it is already centred so this is $\begin{pmatrix}0\\0\end{pmatrix}$.

Now what is the C matrix? If you know, this is $\frac{1}{3}\sum_{i=1}^{3}(x_i - \bar{x})(x_i - \bar{x})^T$, it is the same as $\frac{1}{3}\sum_{i=1}^{3}(x_i\, x_i^T)$. Now this is going to be $\frac{1}{3}\left[\begin{pmatrix}-1\\-1\end{pmatrix}(-1-1) + 0 + \begin{pmatrix}1\\1\end{pmatrix}(1\ 1)\right] = \frac{2}{3}\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$. So, we got C, and what is the next step to do?

Calculate the eigen values, eigen values of C, this turns out to be $\lambda_1 = \frac{4}{3}$, $\lambda_2 = 0$, now corresponding eigen vectors we naturally need only, we can calculate both but I think we need only one of them, u_1 is, let me normalise it, u_2 is $\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$. These are things that leave as homework, I am not going to do this.

Now, what is the, so we need to, what is the best, best 1d space is spanned by u_1 because that is corresponding to the largest eigenvalue which is the same as this, now given this what is the projected data, projected data is given by what is xi tilde if you recall the formula, xi tilde is $(x_i^T u_1)u_1 + (\bar{x_i}^T u_1)u_2 = (x_i^T u_1)u_1$

Now, this is where I am saying $\bar{x} = 0$. So, it just comes down to x_i actually, $(x_i^T u_1)u_1$. So, what is x_1 tilde is , $\frac{1}{2}\left[(-1-1)\begin{pmatrix}1\\1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix}\right] = \begin{pmatrix}-1\\-1\end{pmatrix}$. x_2 tilde is easy to calculate and this is 0. What is x_3 tilde? This is just the $\begin{pmatrix}1\\1\end{pmatrix}$. Now what have you observed? x_1 tilde = x_1, x_2 tilde = x_2, x_3 tilde = x_3. The data is already on a one-dimensional space, then PCA is not doing anything different.

Now, what is the reconstruction error given this observation, this is something that you can directly see, the reconstruction error is just the $J^* = \frac{1}{3}\sum_{i=1}^{3}\left||x_i - \bar{x}|\right|^2 = 0$, So as simple as that, so if the data is already on a line, there is not much work to do, you can directly observe that the given data is already one dimensional. So, you do not have to, so, PCA would not do anything odd, it will just retain the data as it is. Now this was a very simple example.

A summary of today's lecture, which was really tied to the previous lecture in the sense that we started the discussion about PCA in the last lecture. We continued it today and we finally arrived at the PCA algorithm which in a sense is about doing dimensionality reduction by

picking certain eigen values and eigen vectors and using these eigen vectors, you perform projections onto a low dimensional subspace, such that we actually showed that such a scheme minimises the reconstruction error. And we also illustrated these ideas using a very simple example, a two-dimensional example.

In the next lecture, we will look at an alternative equivalent viewpoint of PCA as maximising variance of the data and we will see that the same algorithm can be viewed as maximising variance or minimising the reconstruction error. We have already seen the minimising reconstruction viewpoint in today's and the previous lecture, we will look at the maximising variance viewpoint in the forthcoming lecture.

